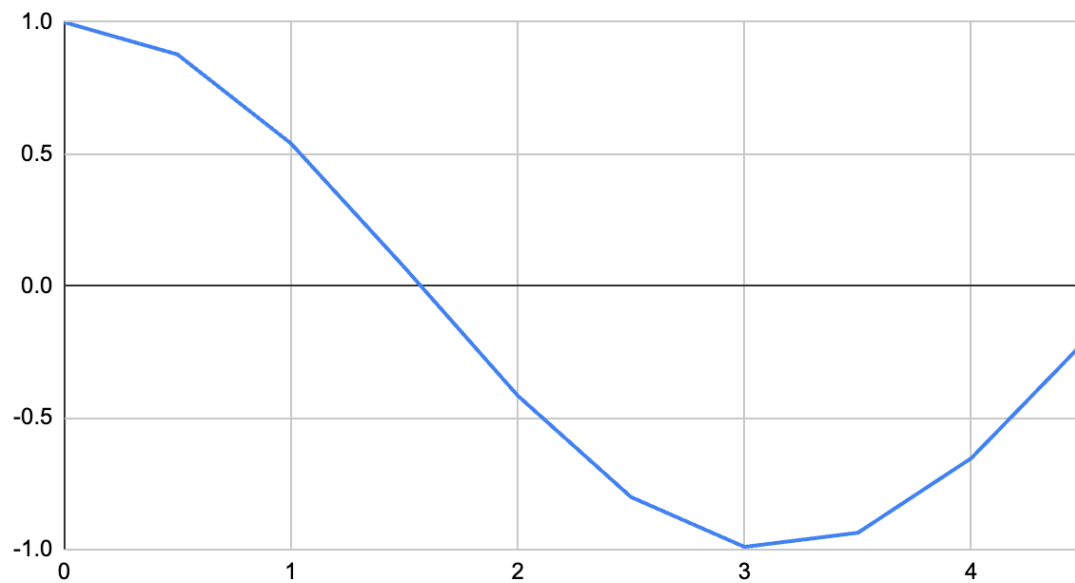


#1.

```
from math import cos
x = [0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5]
for i in range(len(x)):
    print(x[i], "\t", cos(x[i]))
```

x and cosx



Does the shape of your graph resemble the cosine function that you expect? Identify approximately where the function first crosses zero.

- Yes, it looks like a cosine graph. The function crosses zero around $x=1.5$

#2.

```
name = input("Student name: ")
cs_grade = input("Computer Science 1 grade: ")
calc1_grade = input("Calculus I grade: ")
phys1_grade = input("General Physics I grade: ")
eng_grade = input("English Composition grade: ")
sem_grade = input("First-Year Seminar grade: ")

space = 40
data = [["Computer Science 1 grade",cs_grade],
["Calculus I grade",calc1_grade],
["General Physics I grade",phys1_grade],
["English Composition grade",eng_grade],
["First-Year Seminar grade",sem_grade]]

print("\n")
print("Report Card")
print("Student: " + name)
print("\n")
for i in range (0,len(data)):
    character_length = len(data[i][0])
    print(data[i][0] + ": " + "."*(space-character_length) + str(data[i][1]))
```

#3.

```
class_rosters = (  
    ["Computer Science 1", ["Ada Lovelace", "Alan Turing", "Grace  
Hopper"]],  
    ["Calculus I", ["Katherine Johnson", "Isaac Newton", "Emmy Noether"]],  
    ["General Physics I", ["Marie Curie", "Albert Einstein", "Richard  
Feynman"]] )  
print(class_rosters)
```